WORKED SOLUTION

STEP III 2000, Question 6

Pure • Factorising Quartic Polynomials

STEP III 2000, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q6

The key idea

Comparing coefficients turns the factorisation into three equations linking p, q, r to a, b, c. To get an equation for a alone we must eliminate b and c, and the identity $4bc = (b+c)^2 - (b-c)^2$ does exactly that, because $b+c$ and $b-c$ are the combinations that appear naturally. For the final quartic the cubic needs the y^3 term removed first, which a shift $y = x + k$ achieves.

Solution

Finding p, q and r

Expanding the right-hand side,

$$(x^2 - ax + b)(x^2 + ax + c) = x^4 + (b + c - a^2)x^2 + a(b-c)x + bc,$$

where the x^3 terms $ax^3 - ax^3$ cancel. Comparing coefficients,

$$p = b + c - a^2, \qquad q = a(b-c), \qquad r = bc.$$

The cubic for a^2

Write $u = a^2$. From the first equation, $b + c = p + u$.

If $a \neq 0$, the second equation gives $b - c = q/a$, so $(b-c)^2 = q^2/u$. Now

$$4bc = (b+c)^2 - (b-c)^2 \quad \Longrightarrow \quad 4r = (p+u)^2 - \frac{q^2}{u}.$$

Multiplying by $u \neq 0$,

$$4ru = u(u^2 + 2pu + p^2) - q^2 \quad \Longrightarrow \quad u^3 + 2pu^2 + (p^2 - 4r)u - q^2 = 0,$$

so $u = a^2$ is a root.

If $a = 0$, then $q = a(b - c) = 0$, and $u = a^2 = 0$ is a root because every term of the cubic then vanishes at $u = 0$. So in every case a^2 is a root, as required.

Ask yourself Why must there be a root with $u \geq 0$, whatever p, q and r are?

Call the cubic $f(u)$. At $u = 0$ we have $f(0) = -q^2 \leq 0$, and since the leading term is u^3, $f(u) \to +\infty$ as $u \to +\infty$. If $q = 0$ then $u = 0$ is itself a root. Otherwise $f(0) < 0$ and $f(U) > 0$ for some large U, so because f is continuous, the intermediate value theorem gives a root strictly between 0 and U. Either way there is a non-negative root.

This matters because we need $a = \sqrt{u}$ to be real. A non-negative root always gives a real a, so every real quartic with no x^3 term can be split into two real quadratics of the given form. (If $u > 0$ we can recover b and c from $b + c = p + u$ and $b - c = q/\sqrt{u}$.)

The case $p = -1$, $q = -6$, $r = 15$

Here $p^2 - 4r = 1 - 60 = -59$, so the cubic is

$$u^3 - 2u^2 - 59u - 36 = 0.$$

At $u = 9$: $729 - 162 - 531 - 36 = 0$, so $u = 9$ is a root. (Dividing out gives $(u - 9)(u^2 + 7u + 4)$, whose quadratic factor has two negative roots, so $u = 9$ is the only non-negative root.)

Factorising the quartic

The given quartic has a y^3 term, so we shift to remove it. The roots sum to 8, so their mean is 2, and we put $y = x + 2$. Expanding each term:

$$\begin{aligned}
(x+2)^4 &= x^4 + 8x^3 + 24x^2 + 32x + 16,\\
-8(x+2)^3 &= -8x^3 - 48x^2 - 96x - 64,\\
23(x+2)^2 &= 23x^2 + 92x + 92,\\
-34(x+2) &= -34x - 68,
\end{aligned}$$

and adding these and 39,

$$y^4 - 8y^3 + 23y^2 - 34y + 39 = x^4 - x^2 - 6x + 15.$$

This is precisely the case $p = -1$, $q = -6$, $r = 15$, which is why that case appeared.

Take $u = a^2 = 9$ with $a = 3$. Then

$$b + c = p + a^2 = 8, \qquad b - c = \frac{q}{a} = -2,$$

so $b = 3$ and $c = 5$. As a check, $bc = 15 = r$. Hence

$$x^4 - x^2 - 6x + 15 = (x^2 - 3x + 3)(x^2 + 3x + 5).$$

Finally substitute $x = y - 2$:

$$x^2 - 3x + 3 = (y^2 - 4y + 4) - (3y - 6) + 3 = y^2 - 7y + 13,$$
$$x^2 + 3x + 5 = (y^2 - 4y + 4) + (3y - 6) + 5 = y^2 - y + 3.$$

So

$$y^4 - 8y^3 + 23y^2 - 34y + 39 = (y^2 - 7y + 13)(y^2 - y + 3).$$

Taking $a = -3$ instead swaps b and c and gives the same two factors in the other order.

Checks

- Expand the answer: $(y^2 - 7y + 13)(y^2 - y + 3) = y^4 - 8y^3 + (3 + 7 + 13)y^2 - (21 + 13)y + 39$, which matches.
- Put $y = 1$: the quartic is $1 - 8 + 23 - 34 + 39 = 21$, and the factors give $7 \times 3 = 21$.
- Both quadratic factors have negative discriminant (-3 and -11), so neither splits further over the reals, and the quartic has no real roots. That fits $y = 1$ giving a positive value.

We also tested the formulae for p, q, r and the cubic on 2000 random rational choices of a, b, c, and found a non-negative root by bisection for 2000 random (p, q, r).

Common slips

- Dividing by a to get $b - c = q/a$ without dealing with the case $a = 0$.
- Arguing that "a cubic always has a real root", which is true but doesn't show the root is non-negative. The sign of $f(0) = -q^2$ is the point.
- Forgetting to substitute back from x to y at the end.

Beyond STEP

This is essentially Descartes' (and Ferrari's) method for solving the general quartic: shift away the cubic term, then factor into two quadratics by solving an auxiliary cubic, the *resolvent cubic*. Combined with Cardano's formula for cubics, it shows that every quartic can be solved by radicals. Why this stops at degree four, with no such formula for the general quintic, is the subject of Galois theory. Allenby's *Rings, Fields and Groups* is a good place to start.

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